



# CEH v13 — Module 04: Enumeration



## CEH v13 — Module 04

### Enumeration



#### Module Overview

This module covers enumeration techniques for extracting detailed information from target systems through active interaction. Topics include NetBIOS, SNMP, LDAP, NTP, NFS, SMTP, DNS, SMB, RPC, VoIP, and IPsec enumeration, along with defensive countermeasures.

## 1. Enumeration Concepts

### 1.1 What is Enumeration?



#### Definition

Enumeration is the **active** process of extracting detailed information from target systems through direct interaction. Unlike reconnaissance (passive) and scanning (semi-active), enumeration requires established connections and queries to target services.

### Information Gathered Through Enumeration

#### User and Group Information:

- Usernames and account details
- Group memberships and hierarchies

- User privileges and permissions
- Password policies and requirements

**Network Resources:**

- Shared folders and drives
- Network printers and devices
- Available services and applications
- Active sessions and connections

**System Information:**

- Operating system versions and patch levels
- Server banners and service versions
- Routing tables and network topology
- SNMP data and MIB information
- System configuration details

**Policy Information:**

- Security policies and configurations
- Domain policies and trust relationships
- Access control lists (ACLs)
- Authentication mechanisms

## 1.2 Techniques for Enumeration

### Common Enumeration Techniques

**NetBIOS Enumeration**

Extracting Windows networking information via ports 137-139

**SNMP Enumeration**

Querying network device management information

**LDAP Queries**

Enumerating Active Directory users, groups, and organizational units

**SMB Enumeration**

Identifying shares, users, and system information via Server Message Block

### **NFS Exports Enumeration**

Discovering Unix/Linux shared file systems

### **DNS Zone Transfers**

Attempting to retrieve complete DNS zone files

### **SMTP Enumeration**

Validating email addresses using VRFY and EXPN commands

### **SNMP MIB Walking**

Systematically querying Management Information Base objects

### **RPC Enumeration**

Identifying Remote Procedure Call services and endpoints

### **VoIP and SIP Enumeration**

Discovering Voice over IP infrastructure and extensions

### **OS Banner Grabbing**

Collecting service banners revealing software versions

### **IPsec Enumeration**

Identifying VPN endpoints and security associations

### **Active Directory Enumeration**

Comprehensive AD structure and object discovery

## **1.3 Services and Ports to Enumerate**

### **Enumeration Target Reference**

| Service            | Port(s)  | Protocol | Purpose                                |
|--------------------|----------|----------|----------------------------------------|
| <b>DNS</b>         | 53       | TCP/UDP  | Zone transfers, DNS records            |
| <b>SMB/NetBIOS</b> | 139, 445 | TCP      | Users, shares, OS information          |
| <b>SNMP</b>        | 161      | UDP      | MIB information, system details        |
| <b>LDAP</b>        | 389, 636 | TCP      | AD users, groups, organizational units |
| <b>SMTP</b>        | 25, 587  | TCP      | User validation, email enumeration     |

| Service         | Port(s)    | Protocol | Purpose                           |
|-----------------|------------|----------|-----------------------------------|
| <b>RPC</b>      | 135, 111   | TCP      | Service listing, endpoint mapping |
| <b>NTP</b>      | 123        | UDP      | Time synchronization, system info |
| <b>NFS</b>      | 2049       | TCP      | Exported file shares              |
| <b>SIP/VoIP</b> | 5060, 5061 | TCP/UDP  | Extensions, server information    |
| <b>IPsec</b>    | 500, 4500  | UDP      | VPN endpoints, security policies  |
| <b>FTP</b>      | 21         | TCP      | Anonymous access, file listings   |
| <b>SSH</b>      | 22         | TCP      | Supported authentication methods  |
| <b>Telnet</b>   | 23         | TCP      | Banner grabbing, system info      |

## 2. NetBIOS Enumeration



### NetBIOS Overview

NetBIOS (Network Basic Input/Output System) provides information about Windows systems via ports **137 (name service)**, **138 (datagram)**, and **139 (session)**.

### 2.1 NetBIOS Enumeration Tools

#### Built-In Windows Tools

##### nbtstat

Windows built-in NetBIOS statistics tool

```
nbtstat -A <target_ip>
nbtstat -n
```

##### Net Commands

Windows networking commands

```
net view \\<target_ip>
net use
```

```
net user /domain  
net group /domain
```

## Third-Party Tools

### NetBIOS Enumerator

GUI-based NetBIOS information extraction

#### **nbtscan**

Fast NetBIOS name scanner

```
nbtscan 192.168.1.0/24
```

#### **Hyena**

Comprehensive Windows network management tool

#### **enum4linux**

Linux tool for enumerating Windows systems

```
enum4linux -a <target_ip>
```

## 2.2 Enumerating User Accounts

### User Enumeration Methods

#### Domain User Enumeration

```
net user /domain  
net group "Domain Admins" /domain  
net group "Domain Users" /domain
```

#### Local User Enumeration

```
net user  
net localgroup administrators
```

#### Using nbtstat

```
nbtstat -A <target_ip>
```

Reveals:

- Computer name
- Domain/Workgroup
- Logged-in users
- Services running

### Using enum4linux

```
enum4linux -U <target_ip> # Users  
enum4linux -G <target_ip> # Groups  
enum4linux -S <target_ip> # Shares
```

## 2.3 Enumerating Shared Resources Using Net View

### Share Enumeration

#### Basic Net View

```
net view \\<<target_ip>
```

#### Information Extracted:

- Shared folders and drives
- Shared printers
- Administrative shares (C\$, ADMIN\$, IPC\$)
- Hidden shares (ending with \$)

#### Advanced Share Enumeration

```
# Using smbclient  
smbclient -L //<target_ip> -N  
  
# Using crackmapexec  
crackmapexec smb <target_ip> --shares
```

## 2.4 NetBIOS Enumeration Using AI

AI enhances NetBIOS enumeration:

### AI Capabilities

#### Automated Scan Orchestration

Intelligently sequencing NetBIOS enumeration steps

#### Share List Parsing

Automatically categorizing and prioritizing discovered shares

#### Attack Path Mapping

Mapping network names to potential exploitation routes

#### Public SMB Share Detection

Identifying publicly accessible shares with sensitive data

#### Permission Risk Scoring

Scoring share permissions based on access levels and content

#### Correlation Analysis

Linking NetBIOS data with other enumeration findings

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## 3. SNMP Enumeration

### 3.1 Working of SNMP



#### SNMP Architecture

Simple Network Management Protocol (SNMP) is used for managing and monitoring network devices. It operates in a manager-agent model where managers query agents running on network devices.

### SNMP Versions

#### SNMPv1

- Original version
- Plaintext community strings

- No encryption
- Highly insecure

### **SNMPv2c**

- Community-based authentication
- Bulk operations support
- Still uses plaintext
- Backward compatible

### **SNMPv3**

- User-based security model
- Authentication and encryption
- Access control
- Industry standard for secure SNMP

## **Core Components**

### **SNMP Agent**

Runs on managed devices (routers, switches, servers)

### **SNMP Manager**

Queries agents and receives notifications

### **Community Strings**

- **"public"** — Default read-only community string
- **"private"** — Default read-write community string
- Acts as password for SNMP access

### **SNMP Operations:**

- GET — Retrieve specific value
- GETNEXT — Retrieve next value in tree
- GETBULK — Retrieve multiple values
- SET — Modify value (write community required)
- TRAP — Asynchronous notification from agent

## 3.2 Management Information Base (MIB)

**MIB** is a hierarchical database of managed objects identified by Object Identifiers (OIDs).

### Key OID Categories

#### System Information

|                   |                    |
|-------------------|--------------------|
| 1.3.6.1.2.1.1.1.0 | System description |
| 1.3.6.1.2.1.1.3.0 | System uptime      |
| 1.3.6.1.2.1.1.4.0 | System contact     |
| 1.3.6.1.2.1.1.5.0 | System name        |
| 1.3.6.1.2.1.1.6.0 | System location    |

#### Network Interfaces

|                     |                        |
|---------------------|------------------------|
| 1.3.6.1.2.1.2       | Interfaces table       |
| 1.3.6.1.2.1.2.2.1.2 | Interface descriptions |

#### Host Resources

|                  |                    |
|------------------|--------------------|
| 1.3.6.1.2.1.25.1 | System info        |
| 1.3.6.1.2.1.25.2 | Storage            |
| 1.3.6.1.2.1.25.4 | Processes          |
| 1.3.6.1.2.1.25.6 | Software installed |

#### IP Information

|                  |                  |
|------------------|------------------|
| 1.3.6.1.2.1.4    | IP group         |
| 1.3.6.1.2.1.4.20 | IP address table |
| 1.3.6.1.2.1.4.21 | IP routing table |

#### Vendor-Specific

|             |                        |
|-------------|------------------------|
| 1.3.6.1.4.1 | Enterprise/vendor MIBs |
|-------------|------------------------|

## 3.3 Enumerating SNMP Using snmpwalk

**snmpwalk** traverses the entire MIB tree or specific branches.

## Basic Usage

```
# Full MIB walk (SNMPv1)
snmpwalk -v1 -c public <target>

# Full MIB walk (SNMPv2c)
snmpwalk -v2c -c public <target>

# Specific OID walk
snmpwalk -v2c -c public <target> 1.3.6.1.2.1.1

# SNMPv3 (authenticated)
snmpwalk -v3 -u <username> -l authPriv -a SHA -A <auth_pass> -x AES -
X <priv_pass> <target>
```

## Information Revealed

### System Details:

- Hostname and description
- OS version and architecture
- System uptime
- Contact and location information

### Network Configuration:

- IP addresses and subnets
- Network interfaces and MAC addresses
- Routing tables
- ARP cache

### Running Processes:

- Process names and paths
- Running services
- Command-line arguments

**Installed Software:**

- Application names and versions
- Patch levels
- Installed packages

**Storage Information:**

- Disk usage and capacity
- Mount points
- File system types

**User Accounts:**

- Logged-in users (on some systems)
- User account names

## 3.4 SNMP Enumeration Using Nmap

Nmap includes powerful NSE scripts for SNMP enumeration.

### SNMP NSE Scripts

```
# Basic SNMP info
nmap -sU -p 161 --script snmp-info <target>

# SNMP process enumeration
nmap -sU -p 161 --script snmp-processes <target>

# SNMP interface enumeration
nmap -sU -p 161 --script snmp-interfaces <target>

# SNMP NetBIOS name discovery
nmap -sU -p 161 --script snmp-netstat <target>

# SNMP system information
nmap -sU -p 161 --script snmp-sysdescr <target>

# SNMP user enumeration
nmap -sU -p 161 --script snmp-win32-users <target>
```

```
# SNMP community string brute force
nmap -sU -p 161 --script snmp-brute <target>
```

```
# All SNMP scripts
nmap -sU -p 161 --script snmp-* <target>
```

## 3.5 SNMP Enumeration Tools

### Command-Line Tools

#### **snmpwalk**

Traverse MIB trees

#### **snmpget**

Retrieve specific OID values

#### **snmpbulkwalk**

Fast bulk retrieval

#### **onesixtyone**

Fast SNMP scanner and brute forcer

```
onesixtyone -c community.txt <target>
```

#### **snmpcheck**

Comprehensive SNMP enumeration tool

```
snmpcheck -t <target> -c public
```

### GUI Tools

#### **SolarWinds SNMP Walker**

Commercial SNMP browser and analyzer

#### **OpUtils**

Network monitoring and SNMP tool suite

#### **SNMP MIB Browser**

Interactive MIB tree navigation

### **iReasoning MIB Browser**

Free SNMP MIB browser with comprehensive features

## **3.6 SNMP Enumeration with AI**

AI enhances SNMP enumeration efficiency:

### **AI Applications**

#### **Sensitive OID Identification**

Automatically identifying OIDs containing credentials, keys, or sensitive configs

#### **Misconfiguration Detection**

Recognizing SNMP configurations that expose excessive information

#### **Risk Scoring**

Prioritizing SNMP findings based on exploitability and impact

#### **Attack Graph Generation**

Building network topology and attack paths from routing and interface data

#### **Pattern Recognition**

Identifying anomalous SNMP responses indicating deception or honeypots

#### **Automated Reporting**

Generating structured reports with actionable SNMP intelligence

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## **4. LDAP Enumeration**



### **LDAP Overview**

Lightweight Directory Access Protocol (LDAP) is used for accessing and managing directory information services, particularly in Active Directory environments. LDAP enumeration reveals critical organizational structure and user information.

### **4.1 Manual and Automated LDAP Enumeration**

# Manual LDAP Enumeration

## Using Ldapsearch

```
# Anonymous bind (if allowed)
ldapsearch -x -h <target_ip> -b "dc=example,dc=com"

# Authenticated bind
ldapsearch -x -h <target_ip> -D "cn=admin,dc=example,dc=com" -w password -b "dc=example,dc=com"

# Enumerate users
ldapsearch -x -h <target_ip> -b "dc=example,dc=com" "(objectClass=user)"

# Enumerate groups
ldapsearch -x -h <target_ip> -b "dc=example,dc=com" "(objectClass=group)"

# Enumerate computers
ldapsearch -x -h <target_ip> -b "dc=example,dc=com" "(objectClass=computer)"

# Get naming contexts
ldapsearch -x -h <target_ip> -s base namingContexts
```

## Automated LDAP Enumeration

### LDAPDomainDump

```
ldapdomaindump -u 'DOMAIN\user' -p 'password' <target_ip>
```

### Windapsearch

```
# Enumerate users
windapsearch -d example.com --dc <target_ip> -u user@example.com -p password --users

# Enumerate groups
```

```
windapsearch -d example.com --dc <target_ip> -u user@example.com -p  
password --groups
```

```
# Enumerate privileged users
```

```
windapsearch -d example.com --dc <target_ip> -u user@example.com -p  
password --privileged-users
```

## Information Retrieved

### User Information:

- Usernames and display names
- Email addresses
- Last logon timestamps
- Password last set dates
- Account expiration dates
- User attributes and descriptions

### Group Information:

- Group names and descriptions
- Group memberships
- Nested groups
- Privileged groups (Domain Admins, Enterprise Admins)

### Organizational Units (OUs):

- OU structure and hierarchy
- Computer organization
- User organization

### Policy Information:

- Password policies
- Account lockout policies
- Kerberos policies

### Service Principal Names (SPNs):

- Kerberoastable accounts
- Service accounts
- Service mappings

#### **Domain Controllers:**

- DC hostnames
- DC IP addresses
- FSMO role holders

## **4.2 LDAP Enumeration Tools**

### **Command-Line Tools**

#### **ldapsearch**

Standard LDAP query tool (OpenLDAP)

#### **ldapdomaindump**

Dumps Active Directory information in human-readable formats

#### **windapsearch**

Python script for AD enumeration via LDAP

#### **ADEplorer** (Sysinternals)

Windows tool for AD browsing and snapshots

### **GUI Tools**

#### **Softerra LDAP Browser**

Comprehensive LDAP client with query builder

#### **JXplorer**

Open-source LDAP browser

#### **Apache Directory Studio**

Eclipse-based LDAP browser and directory client

### **Nmap LDAP Scripts**

```
# LDAP root DSE enumeration
nmap -p 389 --script ldap-rootdse <target>
```

```
# LDAP search
nmap -p 389 --script ldap-search <target>

# LDAP brute force
nmap -p 389 --script ldap-brute <target>
```

## 5. NTP and NFS Enumeration

### 5.1 NTP Enumeration

**Network Time Protocol (NTP)** can leak sensitive system information.

#### Information Leaked via NTP

##### System Details:

- Operating system version
- System uptime
- Server time and timezone

##### Network Topology:

- NTP peers and stratum levels
- Network time sources
- Server relationships

##### Client Information:

- Recent client connections (via monlist)
- Client IP addresses
- Connection timestamps

### 5.2 NTP Enumeration Commands

#### Using ntpq

```
# Query NTP peers
ntpq -c peers <target>

# Query NTP associations
ntpq -c associations <target>

# Get system variables
ntpq -c sysinfo <target>

# Get detailed peer info
ntpq -c "rv 0" <target>
```

## Using ntpdc (Legacy)

```
# Get monitor list (last 600 clients)
ntpdc -c monlist <target>

# Get system info
ntpdc -c sysinfo <target>

# Get peer info
ntpdc -c peers <target>
```



### Monlist Vulnerability

The `monlist` command reveals the last 600 NTP clients that connected to the server, including their IP addresses. This can be used for:

- Network mapping
- DDoS amplification attacks
- Identifying internal network ranges

Modern NTP implementations disable monlist by default.

## 5.3 NTP Enumeration Tools

### ntpq

Standard NTP query tool

### **ntpd**

Legacy NTP control and monitoring

### **Nmap NTP Scripts**

```
nmap -sU -p 123 --script ntp-info <target>  
nmap -sU -p 123 --script ntp-monlist <target>
```

## **5.4 NFS Enumeration**

**Network File System (NFS)** allows file sharing across networks.

### **Discovering NFS Exports**

```
# Show exported file systems  
showmount -e <target>  
  
# Show all mount points  
showmount -a <target>  
  
# Show directories  
showmount -d <target>
```

### **Mounting NFS Shares**

```
# Create mount point  
mkdir /mnt/nfs  
  
# Mount NFS share  
mount -t nfs <target>:/exported/path /mnt/nfs  
  
# Mount with NFS version 3  
mount -t nfs -o vers=3 <target>:/exported/path /mnt/nfs
```

## **Exploiting NFS Misconfigurations**

### **no\_root\_squash**

Allows root user on client to have root privileges on NFS share

```
# Create SUID binary on NFS share as root
cp /bin/bash /mnt/nfs/bash
chmod +s /mnt/nfs/bash
# On victim, execute: /exported/path/bash -p
```

## 5.5 NFS Enumeration Tools

### showmount

Display NFS exports

### nmap NFS Scripts

```
nmap -p 111,2049 --script nfs-* <target>
nmap -p 111 --script rpcinfo <target>
```

### RPCinfo

```
rpcinfo -p <target>
```

## 6. SMTP and DNS Enumeration

### 6.1 SMTP Enumeration



#### SMTP Enumeration Overview

SMTP (Simple Mail Transfer Protocol) can be exploited to validate user accounts and enumerate email addresses through specific commands.

### SMTP Enumeration Commands

#### VRFY (Verify)

Validates if a user mailbox exists

```
telnet <target> 25
VRFY admin
VRFY root
VRFY user@domain.com
```

### **EXPN (Expand)**

Expands mailing lists and aliases

```
telnet <target> 25
EXPN admin
EXPN all@domain.com
```

### **RCPT TO**

Can indirectly validate email addresses

```
telnet <target> 25
MAIL FROM: test@example.com
RCPT TO: user@target.com
```

## **Response Interpretation**

### **250 Code**

User exists / Command successful

### **550 Code**

User does not exist / Mailbox unavailable

### **252 Code**

Cannot verify user, but will attempt delivery

## **6.2 SMTP Enumeration Using Nmap**

```
# SMTP user enumeration
nmap -p 25 --script smtp-enum-users <target>

# SMTP command support
nmap -p 25 --script smtp-commands <target>
```

```
# SMTP open relay check
nmap -p 25 --script smtp-open-relay <target>

# SMTP NTLM info
nmap -p 25,587 --script smtp-ntlm-info <target>
```

## 6.3 SMTP Enumeration Using Metasploit

### Metasploit SMTP Modules

```
# SMTP user enumeration
use auxiliary/scanner/smtp/smtp_enum
set RHOSTS <target>
set USER_FILE /usr/share/wordlists/users.txt
run

# SMTP version detection
use auxiliary/scanner/smtp/smtp_version
set RHOSTS <target>
run
```

## 6.4 SMTP Enumeration Tools

### smtp-user-enum

Perl script for SMTP user enumeration

```
smtp-user-enum -M VRFY -U users.txt -t <target>
smtp-user-enum -M EXPN -U users.txt -t <target>
smtp-user-enum -M RCPT -U users.txt -t <target>
```

### SWAKS (Swiss Army Knife for SMTP)

Testing and troubleshooting SMTP

```
swaks --to user@target.com --server <target>
```

### Metasploit

Auxiliary SMTP modules

**Nmap**

NSE SMTP scripts

## 6.5 SMTP Enumeration Using AI

AI enhances SMTP enumeration:

### AI Capabilities

#### Banner Anomaly Recognition

Identifying unusual SMTP banners indicating honeypots or deception

#### Username Prediction

Generating likely valid usernames based on discovered patterns

#### Automated Command Replay

Intelligently sequencing SMTP commands for maximum information extraction

#### Response Pattern Analysis

Parsing SMTP responses to identify subtle differences indicating account existence

#### Email Format Inference

Deducing email address formats from partial information

## 6.6 DNS Enumeration



### DNS Enumeration Overview

DNS enumeration extracts domain information including subdomains, mail servers, name servers, and IP addresses. Misconfigurations can expose entire network infrastructures.

### DNS Zone Transfer

#### AXFR (Full Zone Transfer)

Attempts to retrieve complete DNS zone file:

```
# Using dig
dig axfr @nameserver target.com
```

```
# Using host
host -l target.com nameserver
```

```
# Using nslookup
nslookup
server nameserver
ls -d target.com
```

### **If successful, exposes:**

- All subdomains
- IP addresses
- Mail servers
- Name servers
- TXT records (SPF, DKIM, DMARC)
- Service records (SRV)

## **DNS Cache Snooping**

Checks if DNS server has cached specific domains:

```
# Non-recursive query
dig @<dns_server> targetdomain.com A +norecurse
```

```
# If cached, responds immediately
# If not cached, returns SERVFAIL or referral
```

### **Use Cases:**

- Determining visited websites
- Identifying internal domain usage
- Mapping customer/partner relationships

## **DNSSEC Zone Walking**

DNSSEC uses NSEC/NSEC3 records which can leak domain names:

```
# Using ldns-walk
ldns-walk target.com

# Using nsec3walker (for NSEC3)
nsec3walker target.com
```

## DNS Subdomain Enumeration

### Brute Force Methods:

```
# Using dnsenum
dnsenum --enum target.com

# Using fierce
fierce --domain target.com

# Using sublist3r
sublist3r -d target.com
```

## DNS Enumeration Using OWASP Amass

**Amass** is a powerful OSINT and DNS enumeration tool.

```
# Passive enumeration
amass enum -passive -d target.com

# Active enumeration
amass enum -active -d target.com

# Subdomain brute forcing
amass enum -brute -d target.com

# With all techniques
amass enum -d target.com -o output.txt
```

### Capabilities:

- Subdomain discovery via multiple sources
- DNS brute forcing with intelligent wordlists
- API integration (VirusTotal, AlienVault, etc.)
- Correlation with external databases
- DNS relationship graphing
- Recursive subdomain discovery

## DNS Enumeration Using Nmap

```
# DNS brute force
nmap --script dns-brute target.com

# DNS cache snoop
nmap -sU -p 53 --script dns-cache-snoop --script-args 'dns-cache-snoop.mode=timed,dns-cache-snoop.domains={google.com,facebook.com}' <dns_server>

# DNS NSEC enumeration
nmap --script dns-nsec-enum --script-args dns-nsec-enum.domains=target.com

# DNS zone transfer
nmap --script dns-zone-transfer --script-args dns-zone-transfer.domain=target.com <nameserver>

# DNS service discovery
nmap --script dns-srv-enum --script-args dns-srv-enum.domain=target.com
```

## DNS Enumeration with Nmap Using AI

AI enhances DNS enumeration:

### Complex Relationship Mapping

Visualizing intricate DNS hierarchies and dependencies

### Wildcard Detection

Identifying and filtering wildcard DNS records

### **Misconfiguration Identification**

Automatically detecting security weaknesses in DNS configs

### **Exploitable Leak Prioritization**

Ranking DNS findings by exploitability and business impact

## **DNS Cache Snooping Using AI**

AI analyzes cache snooping results:

### **Query Pattern Analysis**

Identifying patterns in cached queries

### **Domain Relationship Mapping**

Connecting related domains and subdomains

### **Internal Domain Discovery**

Inferring internal domain structures from cache behavior

### **Temporal Analysis**

Understanding when and how frequently domains are accessed

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## **7. Other Enumeration Techniques**

### **7.1 IPsec Enumeration**

**IPsec** (Internet Protocol Security) enumeration identifies VPN endpoints and configurations.

#### **IPsec Ports and Protocols**

- **Port 500/UDP** — ISAKMP (Internet Security Association and Key Management Protocol)
- **Port 4500/UDP** — NAT-T (NAT Traversal)
- **Protocol 50** — ESP (Encapsulating Security Payload)
- **Protocol 51** — AH (Authentication Header)

#### **IPsec Enumeration Tools**

## **ike-scan**

Discover and fingerprint IKE hosts

```
# Basic IKE scan
ike-scan <target>

# Aggressive mode scan
ike-scan -A <target>

# PSK cracking
ike-scan -A -M -P <hashfile> <target>
```

## **ikeforce**

Brute force and enumerate IKE VPNs

# **7.2 IPsec Enumeration with AI**

AI assists IPsec enumeration:

### **Weak Cipher Detection**

Identifying outdated or vulnerable encryption algorithms

### **Default Policy Identification**

Recognizing vendor default IPsec configurations

### **VPN Setup Prediction**

Predicting VPN architectures based on port and protocol patterns

### **Configuration Analysis**

Analyzing IKE proposals for security weaknesses

# **7.3 VoIP Enumeration**

**Voice over IP (VoIP)** enumeration discovers telephony infrastructure.

## **SIP Enumeration**

**SIP (Session Initiation Protocol)** uses port **5060 (UDP/TCP)** and **5061 (TLS)**.

## **SIP Enumeration Methods**

### **OPTIONS Request**

```
# Using sipvicious
svmap <target_network>

# Extension enumeration
svwar -e 1000-9999 <target>

# User enumeration
svcrack -u users.txt <target>
```

## ENUM Queries

E.164 number to URI mapping

### Information Gathered:

- SIP server version and capabilities
- Valid extensions
- User accounts
- Supported codecs
- Authentication mechanisms

## 7.4 RPC Enumeration

**Remote Procedure Call (RPC)** enumeration reveals running services.

### RPC Ports

- **Port 135/TCP** — Windows RPC Endpoint Mapper
- **Port 111/TCP** — Unix RPC Portmapper

### RPC Enumeration Tools

#### rpcinfo

```
rpcinfo -p <target>
```

#### rpcclient

```
rpcclient -U "" <target>
> enumdomusers
```

```
> enumdomgroups  
> querydominfo
```

### Nmap RPC Scripts

```
nmap -p 111 --script rpcinfo <target>  
nmap -p 135 --script msrpc-enum <target>
```

## 7.5 Unix/Linux User Enumeration

### Enumeration Methods

#### SSH Banner Grabbing

```
nc <target> 22  
ssh <target>
```

#### Finger Service (Port 79)

```
finger @<target>  
finger username@<target>
```

#### R-Services (Ports 512-514)

Legacy remote execution services

#### rwho (Port 513)

Displays logged-in users

#### rusers

```
rusers <target>
```

### /etc/passwd Exposure

If accessible via LFI or misconfiguration:

```
curl http://target/index.php?page=../../../../etc/passwd
```

## 7.6 SMB Enumeration

**Server Message Block (SMB)** enumeration is crucial for Windows environments.

### SMB Information Gathered

#### User Information:

- Local and domain users
- User SIDs (Security Identifiers)
- User descriptions
- Last logon times

#### Share Information:

- Share names and paths
- Share permissions
- Hidden shares
- Administrative shares

#### System Information:

- Operating system version
- Domain/workgroup name
- Computer name
- SMB signing status

#### Policy Information:

- Password policies
- Lockout policies
- Minimum password length
- Password complexity requirements

### SMB Enumeration Tools

#### **enum4linux**

Comprehensive SMB/NetBIOS enumeration

```
enum4linux -a <target>
```

### **smbclient**

Connect to SMB shares

```
smbclient -L //<target> -N  
smbclient //<target>/share -U username
```

### **crackmapexec**

Advanced SMB enumeration and exploitation

```
crackmapexec smb <target>  
crackmapexec smb <target> --shares  
crackmapexec smb <target> -u user -p password --users  
crackmapexec smb <target> -u user -p password --groups
```

### **smbmap**

SMB share enumeration

```
smbmap -H <target>  
smbmap -H <target> -u username -p password
```

### **Nmap SMB Scripts**

```
nmap --script smb-enum-shares <target>  
nmap --script smb-enum-users <target>  
nmap --script smb-enum-domains <target>  
nmap --script smb-os-discovery <target>  
nmap --script smb-security-mode <target>
```

## **7.7 SMB Enumeration with AI**

AI enhances SMB enumeration:

### **Automated Share Mapping**

Systematically cataloging all accessible shares

## **Permission Analysis**

Analyzing and visualizing complex share permissions

## **Weak ACL Identification**

Detecting misconfigured access control lists

## **Exploitable Share Prioritization**

Ranking shares by likelihood of containing sensitive data

## **Credential Reuse Detection**

Identifying patterns suggesting credential reuse across systems

# **7.8 AI Automation Script for Enumeration**

Comprehensive AI-driven enumeration automation:

## **Script Capabilities**

### **Multi-Protocol Enumeration**

- Simultaneous enumeration across all protocols
- Intelligent protocol detection and adaptation
- Automatic fallback to alternative methods

### **Data Correlation**

- Cross-referencing findings across protocols
- Identifying relationships between discovered assets
- Building comprehensive attack graphs

### **Automatic AD Mapping**

- Complete Active Directory structure discovery
- User and group relationship mapping
- Privilege path identification

### **Finding Prioritization**

- AI-driven risk scoring
- Exploitability assessment
- Business impact evaluation

## Reporting

- JSON/XML structured data export
- HTML interactive reports
- Executive summaries
- Technical detail sections
- Remediation recommendations

# 8. Enumeration Countermeasures



### Defensive Posture

Effective enumeration countermeasures reduce the attack surface by limiting information disclosure and implementing strong access controls.

## 8.1 SNMP Countermeasures

### Protection Strategies

#### Use SNMPv3 with Encryption

Implement authenticated and encrypted SNMP communications

```
# SNMPv3 configuration example
snmp-server group v3group v3 auth
snmp-server user v3user v3group v3 auth sha AuthPass priv aes 128 PrivPass
```

#### Change Default Community Strings

Replace "public" and "private" with complex strings

#### Implement ACLs

Restrict SNMP access to specific management IPs

```
access-list 10 permit 192.168.1.100
snmp-server community SecureString RO 10
```

### **Disable SNMP if Unused**

Remove SNMP service from systems not requiring monitoring

### **Monitor SNMP Traffic**

Log and alert on SNMP queries from unauthorized sources

## **8.2 LDAP Countermeasures**

### **Security Measures**

#### **Disable Anonymous Binds**

Prevent unauthenticated LDAP queries

```
# Active Directory
Set-ADObject -Identity "CN=Directory Service,CN=Windows NT,CN=Services,CN=Configuration,DC=domain,DC=com" -Replace @{dSHeuristics="000002"}
```

#### **Enforce LDAPS (SSL/TLS)**

Encrypt LDAP traffic using port 636

#### **Apply ACL Restrictions**

Limit LDAP query permissions to necessary users only

#### **Harden AD Permissions**

- Remove unnecessary permissions
- Implement least privilege
- Audit permission changes

#### **Enable Signing and Channel Binding**

Prevent LDAP relay attacks

## **8.3 NFS Countermeasures**

### **NFS Hardening**

#### **Restrict Exports to Specific Hosts**

```
# /etc/exports
/shared 192.168.1.0/24(ro,root_squash,no_subtree_check)
```

### **Enable root\_squash**

Map root user to anonymous UID

### **Disable NFS if Not Required**

Remove NFS packages from systems not file sharing

### **Use NFSv4 with Kerberos**

Implement strong authentication

### **Firewall NFS Ports**

Block NFS traffic from public networks

```
iptables -A INPUT -p tcp --dport 2049 -s 192.168.1.0/24 -j ACCEPT
iptables -A INPUT -p tcp --dport 2049 -j DROP
```

## **8.4 SMTP Countermeasures**

### **SMTP Hardening**

#### **Disable VRFY and EXPN Commands**

Postfix:

```
# /etc/postfix/main.cf
disable_vrfy_command = yes
```

Sendmail:

```
# /etc/mail/sendmail.cf
O PrivacyOptions=goaway
```

#### **Enforce Authentication**

Require SMTP AUTH for all connections

#### **Hide Version Information**

Modify SMTP banners to remove version details

### **Implement Rate Limiting**

Limit connections and commands per IP

### **Use SMTP Tarpitting**

Delay responses to suspected enumeration attempts

### **Deploy Greylisting**

Temporarily reject first-time senders

## **8.5 SMB Countermeasures**

### **SMB Security**

#### **Disable SMBv1**

Remove outdated and vulnerable SMB version

```
# Windows
Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol
```

#### **Use Firewall Filtering**

Block SMB from untrusted networks

```
iptables -A INPUT -p tcp --dport 445 -s 10.0.0.0/8 -j ACCEPT
iptables -A INPUT -p tcp --dport 445 -j DROP
```

#### **Require Strong Authentication**

- Disable null sessions
- Enforce NTLMv2 or Kerberos
- Implement SMB signing

#### **Restrict Share Permissions**

- Apply least privilege
- Remove Everyone group permissions
- Audit share access regularly

#### **Block External SMB Traffic**

SMB should never be exposed to the Internet

## 8.6 DNS Enumeration Countermeasures

### DNS Security

#### Restrict Zone Transfers

Limit AXFR to authorized secondary name servers only

```
# BIND configuration
zone "example.com" {
    type master;
    file "/etc/bind/db.example.com";
    allow-transfer { 192.168.1.2; };
};
```

#### Disable Recursive Queries

For authoritative name servers, disable recursion

```
recursion no;
```

#### Implement DNSSEC

Authenticate DNS responses and prevent spoofing

#### Rate Limit DNS Queries

Prevent DNS enumeration and amplification attacks

#### Monitor DNS Query Logs

Detect reconnaissance attempts and anomalous queries

#### Split-Horizon DNS

Provide different responses to internal vs. external queries

---



### Module Summary

This module covered comprehensive enumeration techniques across multiple protocols and services. Key takeaways include:

- **Active Nature:** Enumeration requires direct interaction with target systems
- **Protocol Diversity:** Different protocols reveal different types of information
- **Tool Proficiency:** Mastering various enumeration tools is essential
- **AI Enhancement:** AI significantly improves enumeration efficiency and analysis
- **Defense:** Implementing proper countermeasures reduces enumeration effectiveness

Enumeration bridges the gap between scanning and exploitation, providing the detailed information necessary for successful penetration testing. Understanding both offensive enumeration techniques and defensive countermeasures is critical for comprehensive security assessment.